

Over-Refusal of Benign Slovene Prompts in Gemma-3-12B Is a Criterion Shift

Abstract

Safety-aligned language models sometimes refuse benign requests, and this over-refusal can differ across languages. We apply signal-detection theory (SDT) to a controlled ablation study comparing Gemma-3-12B-IT with its Slovene-adapted sibling GaMS3-12B-Instruct. On the reserved HARD evaluation set (1,093 safe and 349 harmful items scored by a local Qwen3-14B judge whose readout gate never passed), unedited Gemma false-refuses benign Slovene items at a higher rate than English items (0.569 vs 0.364, 1.56 times higher), with GaMS3 at 0.273 in both languages. SDT decomposition attributes the gap to a criterion shift: a difference-in-differences of +0.52 [0.38, 0.68] on the decision threshold, with no corresponding change in sensitivity. The finding is a single exploratory estimate: 8 of 9 compliance-valid rows across five analyses point the same way as the headline (a post-hoc count), but the sign reverses under partial-as-refusal coding. A secondary analysis of reply-language refusal yields a Gemma-specific output-language effect, also rated ESTIMATE due to weak judge anchoring (harmful-content adjudicator $\kappa = 0.12$). These measurements indicate that multilingual safety evaluations should report false-alarm rates on benign items alongside attack-success rates.

1 Introduction

Safety alignment in large language models targets harmful-request refusal, but the same training can produce over-refusal: the model declines benign requests it should answer [Röttger et al., 2023, Souly et al., 2024]. Over-refusal degrades usefulness, and prior work has shown that refusal rates differ across languages [Shen et al., 2024, Yong et al., 2023, Wang et al., 2023, Deng et al., 2023]. Whether these differences reflect a sensitivity gap (the model is less able to distinguish harmful from benign items) or a criterion shift (the model applies a stricter threshold at the same discriminative ability) has rarely been tested. Signal-detection theory (SDT) provides the decomposition: sensitivity d' measures how well the model separates harmful from benign items, while criterion c measures where it sets the decision threshold [Aziz et al., 2026].

Abliteration, the directional ablation of the refusal direction in a model’s residual stream [Arditi et al., 2024], provides a way to study refusal at varying strengths. We compare two 12-billion-parameter models: Gemma-3-12B-IT and GaMS3-12B-Instruct, a Slovene-adapted variant built on the same Gemma-3 base [Vres et al., 2026]. The ablation edit is applied via per-layer LoRA adapters [Hu et al., 2021] optimized by TPE [Akiba et al., 2019].

The study addresses two questions. First, does the model-language interaction on unedited refusal reflect a criterion shift or a sensitivity difference? Second, does the residual refusal after a partial English edit track the language of the model’s reply or the language of the prompt?

Summary of Contributions

1. **Over-refusal as criterion shift (Section 4.1).** Unedited Gemma false-refuses benign Slovene items at 0.569 vs 0.364 in English; GaMS3 stays at 0.273 in both. The criterion

difference-in-differences is $+0.52$ $[0.38, 0.68]$ with no sensitivity change (ESTIMATE; sign reverses under partial-as-refusal coding).

2. **Reply-language refusal (Section 4.2).** A crossed 3×3 design yields a raw output-language effect for Gemma, but PPI++ correction and robustness checks rate it ESTIMATE with a weak anchor.
3. **Measurement methodology (Section 5).** We report how judge specificity, language compliance, and correction methods interact when evaluating multilingual refusal.

2 Related Work

Multilingual refusal gaps. Several studies document that safety alignment degrades in non-English languages, enabling jailbreaks via low-resource languages [Yong et al., 2023, Deng et al., 2023, Shen et al., 2024, Wang et al., 2023]. Recent work separates input-language from output-language effects [Marchisio et al., 2024, Zhang et al., 2026] and shows that safety pathways are shared across languages [Miao et al., 2026, Upadhyaya and Sikdar, 2026, Wang et al., 2025]. The question of whether multilingual refusal gaps reflect over-refusal on benign items rather than under-refusal on harmful items is partially addressed by Addagada [2026], who reports a relevance curse in non-English output, and by Nguyen et al. [2026], who cross prompt and response language on benign content. Yoon et al. [2026] document the disparate cost of safety alignment across languages. The novelty of the present work is partially occupied by these concurrent findings; our contribution is the signal-detection decomposition and the controlled sibling-pair design.

Abliteration. Ardit et al. [2024] showed that refusal in LLMs is mediated by a single direction in the residual stream, and that ablating this direction removes refusal. Joad et al. [2026] demonstrated that multiple directions contribute. Wang et al. [2025] found that the refusal direction generalizes across safety-aligned languages. Krasnodebska et al. [2026] found that English-only DPO alignment does not ensure cross-lingual safety. Hawkins et al. [2026] examined the safety impacts of benign multilingual fine-tuning.

Evaluation methodology. LLM-as-judge evaluations carry systematic biases that compound across languages [Lee et al., 2025]. Prediction-powered inference (PPI++) provides a framework for correcting classifier estimates against a gold standard [Angelopoulos et al., 2023]. Fiedler [2026] shows that the Rogan–Gladden correction is unstable at low Youden J . StrongREJECT [Souly et al., 2024] introduced rubric-based evaluation of jailbreak quality. Aziz et al. [2026] showed that low-resource safety failures are calibration failures, not representation failures.

3 Method

3.1 Models and Edit Procedure

We study Gemma-3-12B-IT and GaMS3-12B-Instruct, both loaded in NF4 (4-bit normal-float) quantization. GaMS3 extends Gemma-3 with Slovene continued pretraining on 140 billion tokens and supervised fine-tuning on a mix that is roughly 79% Slovene [Vres et al., 2026].

The ablation edit applies per-layer directional ablation via LoRA adapters [Hu et al., 2021], optimized by a 40-trial TPE search [Akiba et al., 2019]. Each trial selects a layer subset, computes a refusal direction from contrastive pairs of harmful and benign English prompts, and projects it out

of the residual stream. The edit objective is to reduce English refusal while preserving capability on six benchmarks.

We use two edit doses throughout: a low dose ($\lambda = 0.5$ for both models) and a high dose ($\lambda = 1.0$ for Gemma, 0.625 for GaMS3), chosen to bracket the 50% English refusal point.

3.2 Evaluation Design

Over-refusal (criterion shift). Experiment 5 tests 1,093 safe items (XSTest [Röttger et al., 2023] and OR-Bench) and 349 harmful items in English and Slovene. The Slovene items are machine-translated with Gemini (gemini-2.5-flash); English items are back-translations. Each item is classified as refused or complied by a local Qwen3-14B judge, calibrated against gemini-2.5-flash on DEV (binary κ 0.68–0.86 across cells). The hit rate (refusal on harmful items) and false-alarm rate (refusal on safe items) are computed per model $\times$ language cell, and signal-detection parameters d' and c are derived. The key contrast is the difference-in-differences (DiD), which isolates the model-by-language interaction:

$$\text{DiD}_c = (c_{\text{GaMS3,SL}} - c_{\text{GaMS3,EN}}) - (c_{\text{Gemma,SL}} - c_{\text{Gemma,EN}}). \quad (1)$$

A positive DiD_c means Gemma’s criterion shifts more toward refusal in Slovene than GaMS3’s does.

Crossed grid (reply-language refusal). Experiment 14 crosses three prompt languages (English, Slovene, Hungarian) with three reply languages (English, Slovene, Hungarian) using a parallel one-line suffix in the input language. Each cell contains 200 harmful items and 100 benign items at three dose levels (zero, low, high). The output-language mechanism contrast OUT_{SL} is the mean log-odds refusal gap between Slovene-output and English-output cells, averaging over prompt languages. The input-language contrast IN_{SL} is defined symmetrically.

Judges and corrections. The primary judge for Experiment 14 is validated against blind author-model (not human) adjudication on 240 pooled edited rows: sensitivity = 0.89, specificity = 0.73, $\kappa = 0.51$. In the decisive Gemma Slovene-edited cell ($n = 60$), specificity drops to 0.42. PPI++ correction [Angelopoulos et al., 2023] is reported alongside raw estimates. Experiment 16 uses a held-out 294-item body with PPI as the primary estimator. The harmful-content adjudicator κ in Experiment 16 is 0.12, pooled over $n = 400$, well below the 0.60 gate.

Language compliance. Cells where fewer than 90% of model replies match the requested output language are excluded from mechanism claims or marked. The excluded cells for GaMS3 are: SL $\rightarrow$ EN (compliance 0.00), HU $\rightarrow$ EN (0.14), SL $\rightarrow$ HU (0.56), and HU $\rightarrow$ SL (0.86).

3.3 Hungarian as a Control Language

Hungarian serves as a typologically distant control. The minimum detectable effect for Hungarian contrasts is approximately 3 log-odds units, far larger than any observed effect; all Hungarian results are uninformative.

4 Results

4.1 Over-Refusal: Criterion Shift on Benign Items (ESTIMATE)

The main measurement is that Gemma applies a stricter refusal criterion in Slovene than in English on benign items, while GaMS3 does not. Table 1 reports the signal-detection parameters from Experiment 5.

Table 1: Signal-detection parameters by model and language (Experiment 5, 1,093 safe + 349 harmful items, Qwen3-14B judge).

	Gemma EN	Gemma SL	GaMS3 EN	GaMS3 SL
Hit rate (H)	0.89	0.94	0.90	0.86
False-alarm rate (F)	0.364	0.569	0.273	0.273
d'	1.57	1.39	1.86	1.68
c	-0.44	-0.87	-0.33	-0.23

The DiD on the criterion parameter is $+0.52$ $[0.38, 0.68]$ (95% CI). Gemma’s Slovene criterion is shifted by half a standard deviation relative to the GaMS3 baseline. The DiD on d' is -0.003 $[-0.33, 0.29]$, indistinguishable from zero: the asymmetry is in the threshold, not the sensitivity. Figure 1 shows the decomposition.

Rating and caveats. We use a three-level evidence rating: ROBUST (survives all pre-registered checks), ESTIMATE (the point estimate is informative but at least one sensitivity check fails), and INCONCLUSIVE (the confidence interval includes zero or the result is fragile). This finding is rated ESTIMATE for three reasons. First, the sign reverses to -0.31 when partial answers are coded as refusals rather than compliance. Second, 8 of 9 compliance-valid rows across five analyses (Experiments 5, 9, 14, 15, and a secondary rescoring) point the same way as the headline, but this count was a post-hoc refinement made after the frozen rule found an unresolved cross-body disagreement. Third, a Qwen3-14B rescoring gives a similar interval $[0.39, 0.67]$, but this is a re-derivation from the same Experiment 5 items, not an independent sample.

Edited-model criterion shift. On edited models, the criterion difference is inconclusive: Experiment 11 HARD post-edit $\text{DiD}_c = 0.14$ $[-0.78, 1.12]$, whose CI includes zero.

4.2 Reply-Language Refusal (ESTIMATE, Weak Anchor)

In the 3×3 crossed grid (Experiment 14), edited Gemma at high dose shows a pattern in which residual refusal is higher when the model replies in Slovene than in English. Tables 2 and 3 give the refusal rates; Figure 2 visualizes the full grid.

Table 2: Refusal rates at high edit dose, Gemma (harmful items, 200 per cell).

	EN out	SL out	HU out
EN in	0.18	0.77	0.46
SL in	0.07	0.56	0.53
HU in	0.10	0.83	0.55

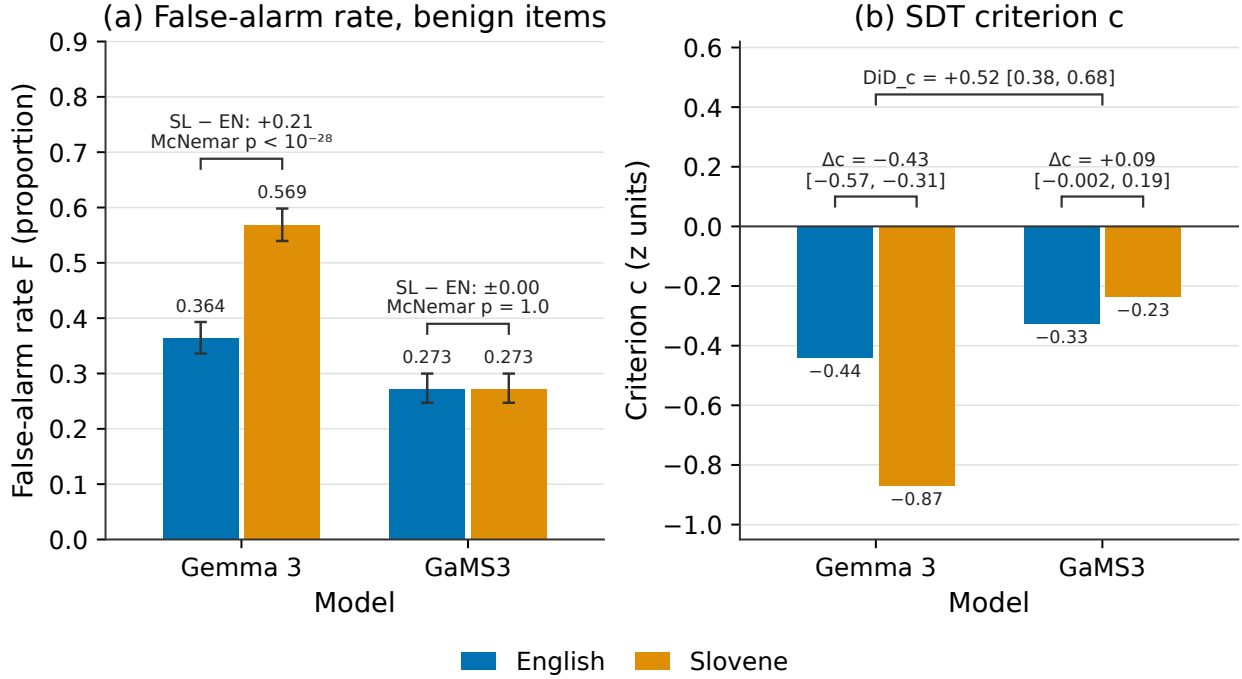


Figure 1: Signal-detection parameters from Experiment 5 on the HARD set (1,093 benign and 349 harmful items; Slovene machine translations vs English back-translations; refusals labelled by a local Qwen3-14B judge, with partial answers coded as non-refusals). Blue bars are English prompts and orange bars are Slovene prompts, for Gemma-3-12B-IT (“Gemma 3”) and GaMS3-12B-Instruct (“GaMS3”). (a) False-alarm rate F , the proportion of benign items refused, with Wilson 95% confidence intervals. Brackets give the paired Slovene–English difference and an exact McNemar test on the same items. Gemma’s rate rises from 0.364 to 0.569 ($p < 10^{-28}$), while GaMS3 refuses 0.273 of benign items in both languages ($p = 1.0$). (b) SDT criterion c in z units, where more negative means a greater tendency to refuse. Lower brackets show the within-model shift Δc (Slovene–English, 95% CI): Gemma moves toward refusing in Slovene ($c = -0.87$ vs -0.44 ; $\Delta c = -0.43$ $[-0.57, -0.31]$), while GaMS3 does not (-0.23 vs -0.33 ; $\Delta c = +0.09$ $[-0.002, 0.19]$). The upper bracket gives the difference-in-differences $\text{DiD}_c = +0.52$ $[0.38, 0.68]$. The corresponding DiD on sensitivity d' (-0.003 $[-0.33, 0.29]$) is indistinguishable from zero, so the asymmetry lies in the threshold rather than in discrimination. This finding is rated ESTIMATE: coding partial answers as refusals reverses DiD_c to -0.31 $[-0.62, -0.09]$.

Table 3: Refusal rates at high edit dose, GaMS3 (harmful items, 200 per cell; cells below 90% compliance marked †).

	EN out	SL out	HU out
EN in	0.21	0.14	0.23
SL in	0.16†	0.15	0.17†
HU in	0.19†	0.12†	0.28

The raw mechanism contrast is $\text{OUT}_{\text{SL}} = +1.14 [0.60, 1.56]$ log-odds for Gemma and the input-language contrast is $\text{IN}_{\text{SL}} = -0.60 [-1.02, -0.18]$. For GaMS3, the input-language contrast $\text{IN}_{\text{SL}} = -0.86 [-1.58, -0.13]$ excludes zero. The cross-model output contrast is $\text{dOUT}_{\text{SL}} = -1.58 [-2.22, -0.66]$. However, the PPI-corrected output contrast for Gemma is $0.20 [-0.76, 1.13]$, consistent with zero, and the robustness check across readouts does not pass.

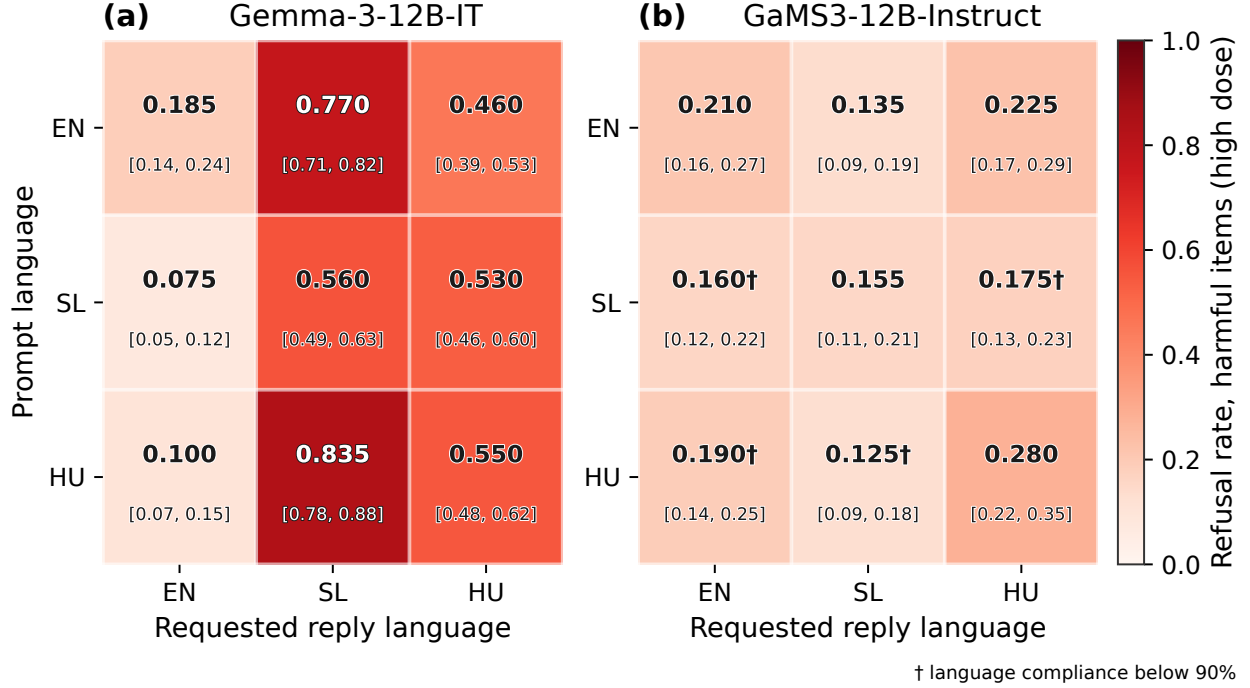


Figure 2: Refusal rates on harmful items at high edit dose in the 3×3 prompt-language $\times$ requested-reply-language grid (Experiment 14, 200 harmful items per cell, primary-judge raw readout). Rows give the prompt language and columns the requested reply language (EN, SL, HU). Colour encodes the refusal rate on a shared white-to-dark-red scale from 0 to 1. Each cell prints the rate with its Wilson 95% CI beneath. (a) Gemma-3-12B-IT: the Slovene-reply column is the most refused (0.560–0.835), the English-reply column the least (0.075–0.185), and the Hungarian-reply column sits in between (0.460–0.550). (b) GaMS3-12B-Instruct: refusal is uniformly low (0.125–0.280) with no reply-language column pattern. † marks cells whose language compliance, pooled over the three dose levels, is below 90%. All four are GaMS3 cells off the English-prompt row (SL→EN, SL→HU, HU→EN, HU→SL), so their reply-language labels are unreliable. The Gemma output-language contrast holds on this raw readout, but its PPI-corrected estimate includes zero. The CIs inside the darkest Gemma cells (SL-reply column, top to bottom) are [0.71, 0.82], [0.49, 0.63], [0.78, 0.88].

Experiment 16 confirmation. On a fresh 294-item body, Experiment 16 gives the PPI-corrected estimate $\text{OUT}_{\text{Gemma}} = +2.43 [+0.89, +5.38]$ log-odds, and the cross-model difference $\text{dOUT} = -2.31 [-5.34, -0.68]$ excludes zero. All confirmation clauses pass on the consensus reference. However, the verdict is capped at ESTIMATE because the harmful-content adjudicator κ is 0.12 (pooled over $n = 400$), well below the 0.60 gate. The adjudicator disagreement stems from truncation: one judge scored cut-off partial compliance as non-harmful, while the other graded the partial content 2–4. The anchor is weak.

SDT decomposition of the output-language effect. At high dose, the Slovene-output criterion DiD (Gemma minus GaMS3) is -1.56 $[-1.97, -1.29]$ and the d' DiD is 0.72 $[-0.06, 1.30]$, which includes zero. This is a separate reply-language criterion contrast, computed at the cell level as Gemma minus GaMS3; after harmonising the sign convention, it is counted among the compliance-valid rows that agree with the headline.

Compliant-row contrast. Restricting to rows where both models achieve $>90\%$ compliance (EN-input rows only), the edit-induced cross-model contrast is $G3_{\text{edit}}(\text{EN} \rightarrow \text{SL}) = -1.26$ $[-2.17, 0.20]$, whose CI includes zero.

Benign false refusal. From blind adjudication ($n = 30$ per cell), Gemma falsely refuses benign $\text{EN} \rightarrow \text{SL}$ prompts at 0.20 $[0.10, 0.37]$ vs $\text{EN} \rightarrow \text{EN}$ at 0.10 $[0.03, 0.26]$.

Refusal composition. Among Gemma’s high-dose $\text{EN} \rightarrow \text{SL}$ harmful-item responses (blind adjudication, $n = 50$), 54% are explicit refusals and 26% are deflections. The deflection rate, defined as the proportion of non-refusal responses that are off-topic deflections rather than substantive answers, differs between Gemma $\text{EN} \rightarrow \text{SL}$ and $\text{EN} \rightarrow \text{EN}$ by 0.46 $[0.23, 0.65]$, consistent with the relevance curse described by Shen et al. [2024] and the output-language effect reported by Addagada [2026].

4.3 Dose-Response

Figure 3 plots refusal rate against ablation dose for both models. Experiment 9 measures refusal rates across nine λ values for both models in English and Slovene (300 items per model $\times$ language cell at $\lambda = 0$, 100 items at each edited step). Gemma’s dose-response curve in Slovene is right-shifted relative to its English curve: at matched doses, more Slovene refusal remains. GaMS3 shows a smaller shift. The fine-grained 13-step dose ladder (Experiment 15) gives the most powerful estimate: $G3 = -0.70$ $[-1.01, -0.42]$ with a minimum detectable effect of 0.42 . The edit-induced component is $+0.47$ $[-0.96, 2.14]$, whose CI includes zero; whether the cross-model gap is baseline or edit-induced is inconclusive.

4.4 Inconclusive and Null Results

Unedited cross-model gap at Slovene-to-Slovene prompts. Experiment 15 gives $G3_{\text{orig}} = -1.17$ $[-2.83, 0.22]$. This is a ceiling artefact: its fragility index, the number of item reclassifications needed to move the CI boundary past zero, is 1 (a single item reclassification in the Gemma $\lambda=0$ Slovene cell, where 299 of 300 items are refused, moves it inside the margin). Only the Experiment 9 body has a robust offset (fragility index 9).

Hungarian. All Hungarian contrasts are uninformative. The minimum detectable effect is approximately 3 log-odds units.

Experiment 16 zero-dose. The Experiment 16 zero-dose Gemma criterion DiD is -0.65 $[-0.90, -0.46]$, rated ESTIMATE. It does not confirm the Experiment 5 headline because it uses a different item body, a different judge, and a suffix-based output-language design rather than paired translation.

RefusEU validation. Experiment 17 tests attack success rate on 1,300 EN + 1,300 SL RefusEU prompts. The edit is effective in all four model $\times$ language cells (ROBUST), with ASR deltas of 0.657–0.937. The EN-vs-SL within-model difference is not interpretable as a language effect because the prompt sets are different natural prompts (LaBSE similarity 0.55).

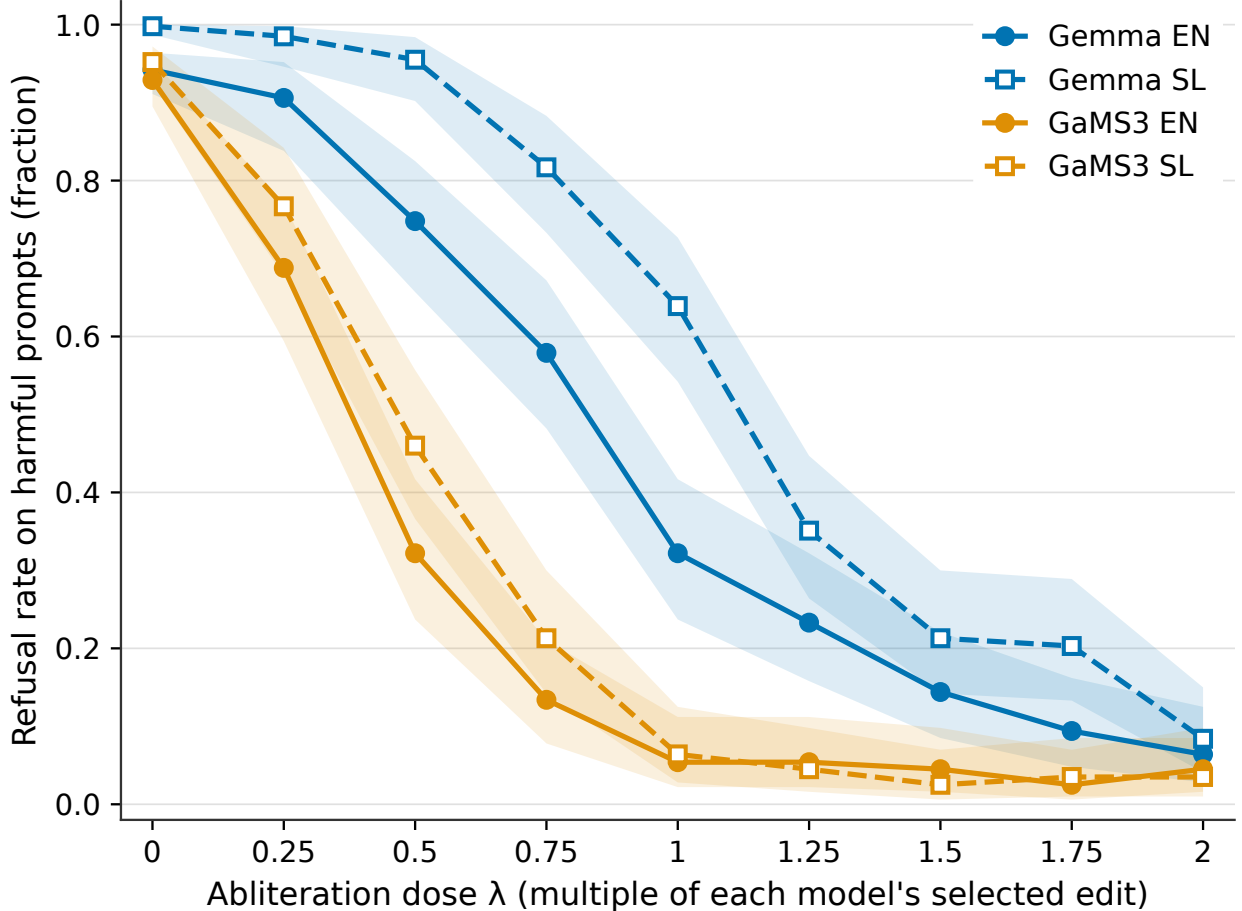


Figure 3: Refusal rate on harmful prompts as a function of ablation dose λ for Gemma-3-12B-IT (blue) and GaMS3-12B-Instruct (amber), in English (solid lines, filled circles; EN-BT arm) and Slovene (dashed lines, hollow squares; item-matched SL-MT arm). λ scales each model’s own selected ablation edit (9 steps from 0 to 2; $\lambda = 1$ is the selected edit), so equal λ does not mean an equal edit across models. Points are J1-judged refusal rates from the Experiment 9 RefusEU-TRAIN probe, with the Hautus correction $(k + 0.5)/(n + 1)$ used in the Experiment 9 analysis. Shaded bands are Wilson 95% intervals on the raw proportion. $n = 300$ harmful items at $\lambda = 0$ and the 100-item core block at every $\lambda > 0$. Gemma’s Slovene curve lies to the right of its English curve throughout the transition (0.64 vs. 0.32 at $\lambda = 1$): at each dose, more Slovene refusal remains. GaMS3’s Slovene curve lies only slightly above its English curve (largest gap 0.14, at $\lambda = 0.5$), and the two meet near the floor from $\lambda = 1$ onward. These per-dose gaps are descriptive. The pre-registered cross-model comparison is made at matched English refusal, not at matched dose.

5 Discussion

5.1 Interpretation

The criterion-shift measurement has a concrete interpretation for multilingual evaluation: a model can appear “safer” in a non-English language because it refuses more benign requests, inflating the refusal rate without any improvement in harm detection. The SDT framing separates these two components. The near-zero DiD on d' means that the language shift does not make either model better or worse at distinguishing harmful from benign items; the measured difference is in the threshold. Aziz et al. [2026] described a similar calibration-not-representation pattern across 23 languages; the present sibling-pair design adds GaMS3-specific detail but does not change the interpretation.

The reply-language finding, if confirmed by future work with validated judge anchors, would suggest that Gemma’s refusal mechanism interacts with output-language processing. Upadhyaya and Sikdar [2026] showed that safety features are geometrically entangled with language-identity features; the present study’s 3×3 design separates the input and output dimensions but cannot determine the mechanism.

5.2 Limitations

ESTIMATE rating on the primary finding. The criterion-shift DiD reverses sign (-0.31) when partial answers are coded as refusals. The choice of coding rule therefore determines whether over-refusal is attributed to Gemma or is absent. This fragility bounds how strongly the finding can be stated. Furthermore, restricting Experiment 5 to the XSTest items alone gives $\text{DiD}_c = 0.10$ $[-0.13, 0.35]$, so the effect is carried by the OR-Bench items.

Judge quality. The Qwen3-14B judge for Experiment 5 was calibrated against gemini-2.5-flash on DEV (κ 0.68–0.86) but not against human labels. On edited rows across the study, no judge exceeds $\kappa = 0.51$ against blind adjudication. The readout gate has never passed. All adjudication in this study is author-model LLM, not human review.

Sample size. The adjudication sample is 30 items per cell for benign false-refusal rates, limiting precision. The refusal-composition numbers (54% explicit, 26% deflection) come from $n = 50$ high-dose harmful items.

Language compliance. Four GaMS3 cells fall below 90% compliance (SL→EN at 0.00, HU→EN at 0.14, SL→HU at 0.56, HU→SL at 0.86). GaMS3 did not follow the output-language instruction off the English-input row; its SL→EN responses were written in Slovene, not English. Mechanism claims for GaMS3 are therefore restricted to the compliant EN-input row.

Quantization. Both models run in NF4 (4-bit) quantization, which may affect refusal behaviour relative to full-precision weights.

Novelty. The observation that over-refusal rates differ across languages is partially anticipated by Addagada [2026] and Nguyen et al. [2026]. The present work’s contribution is the signal-detection decomposition and the controlled sibling-pair design.

6 Conclusion

Unedited Gemma-3-12B-IT false-refuses benign Slovene items at 0.569 vs 0.364 in English. Signal-detection analysis attributes this to a criterion shift ($\text{DiD}_c = +0.52$ $[0.38, 0.68]$) with no sensitivity difference. The finding is rated ESTIMATE because the sign reverses under an alternative coding rule. On edited models, the criterion difference is inconclusive (0.14 $[-0.78, 1.12]$). A secondary analysis of reply-language refusal yields a PPI-corrected output-language effect of +2.43 $[+0.89, +5.38]$ for Gemma, but this is also rated ESTIMATE because the harmful-content adjudicator κ is 0.12. Multilingual safety evaluations that report only attack-success rates on harmful items will miss criterion shifts that inflate refusal rates on benign items.

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